

The chemical industry

1. recall and use the terms 'bulk' (made on a large scale) and 'fine' (made on a small scale) in terms of the chemical industry;
2. recall examples of chemicals made on a large scale (ammonia, sulfuric acid, sodium hydroxide, phosphoric acid) and examples of chemicals made on a small scale (drugs, food additives, fragrances);
3. interpret information about the work done by people who make chemicals (no recall expected);
4. understand that new chemical products or processes are the result of an extensive programme of research and development (for example catalysts for new processes);
5. understand that governments have strict regulations to control chemical processes as well as the storage and transport of chemicals to protect people and the environment.

What are the characteristics of green chemistry?

6. understand that the production of useful chemicals involves several stages (to include the preparation of feedstocks, synthesis, separation of products, handling of by-products and wastes, and the monitoring of purity);
7. understand that sustainability of any chemical process (laS 6.2) depends on:
 - whether or not the feedstock is renewable;
 - the atom economy;
 - the nature and amount of by-products or wastes and what happens to them;
 - the energy inputs or outputs;
 - the environmental impact;
 - the health and safety risks (laS 5.3 & 5.4);
 - the social and economic benefits.
8. understand that a catalyst provides an alternative route for a reaction with a lower activation energy;
9. **represent chemical reactions by balanced equations, including state symbols;**
10. **calculate the masses of reactants and products from balanced equations including state symbols;**
11. **calculate the masses of reactants and products from balanced equations.**

Making ethanol by three methods

12. understand how ethanol is made on an industrial scale as a fuel, a solvent and as a feedstock for other processes;
13. understand that there is a limit to the concentration of ethanol solution that can be made by fermentation;
14. understand how ethanol solution can be concentrated by distillation to make products such as whisky and brandy;
15. understand the optimum conditions for making ethanol by fermentation of sugar with yeast, taking into consideration temperature and pH;
16. understand how genetically modified E coli bacteria can be used to convert waste biomass from a range of sources into ethanol and recall the optimum conditions for the process;
17. recall in outline the synthetic route for converting ethane (from oil refining) into ethanol;
18. interpret data about these processes and evaluate their sustainability (laS 6.5).

4.1 Units of Assessment

GCSE Chemistry A (J634)

Unit 1: Chemistry A Unit 1 – modules C1, C2, C3 (A321)

16.7% of the total GCSE marks
40 minutes written paper
42 marks

This question paper:

- is offered in Foundation and Higher Tiers;
- focuses on modules C1, C2 and C3;
- uses objective style questions throughout (there is no choice of questions).
- assesses knowledge and understanding of the specification content and application of that knowledge and understanding.

Unit 2: Chemistry A Unit 2 – C4, C5, C6 (A322)

16.7% of the total GCSE marks
40 minutes written paper
42 marks

This question paper:

- is offered in Foundation and Higher Tiers;
- focuses on modules C4, C5 and C6;
- uses objective style questions throughout (there is no choice of questions);
- assesses knowledge and understanding of the specification content and application of that knowledge and understanding.

Unit 3: Chemistry A Unit 3 – Ideas in Context plus C7 (A323)

33.3% of the total GCSE marks
1 hr written paper
55 marks

This question paper:

- is offered in Foundation and Higher Tiers;
- assesses knowledge and understanding of the specification content and application of that knowledge and understanding;
- incorporates pre-release material;
- the subject focus of the pre-release material will normally be one or two of modules C1 to C6, the precise focus will be clear from the stimulus material;
- the remaining questions will be focused on the content of C7 Further Chemistry;
- uses structured questions throughout (there is no choice of questions);
- includes some marks for communication skills.

33.3% of the total GCSE marks
skills assessment
40 marks (16 + 24)

- This skills assessment unit comprises two elements: the critical analysis of primary data and a case study of a topical (scientific) issue.
- Opportunities for both elements should arise naturally during the course.
- This unit is assessed by teachers, internally standardised and then externally moderated by OCR.

Unit 5: Chemistry A Unit 5 – Practical Investigation (A330)

33.3% of the total GCSE marks
skills assessment
40 marks

- This unit comprises five strands, which together are used to assess a complete investigative task.
- This unit is assessed by teachers, internally standardised and then externally moderated by OCR.

Options

Units 1, 2 and 3 **and** either Unit 4 **or** Unit 5.

are set in one of two tiers: Foundation Tier and Higher Tier. Foundation Tier papers assess Grades G to C and Higher Tier papers assess Grades D to A*. Candidates are entered for either the Foundation Tier or the Higher Tier using option codes F and H.

Skills assessment) are not tiered. Candidates enter either A329, Practical Investigation, or A330, Practical Investigation.

Candidates can enter Units 1, 2, and 3 at different tiers, so for example, a candidate can enter Unit 1 at Foundation Tier and A323H.

Assessment Availability

Examination sessions each year, in January and June.

Unit 1 (A321)	Unit 2 (A322)	Unit 3 (A323)	Unit 4 (A329)
–	–	–	–
✓	–	–	–
✓	✓	–	–
✓	✓	✓	✓

Units 1, 2, 3, A321 and A322 will be available in the January and June sessions. Units 4, 5, A323 and skills assessment (Units A329 and A330) will be available in the January and June sessions.

Foundation and Higher tier papers covering the same unit will be timetabled or co-timed at the same time. The papers timetabled simultaneously will cover different questions, targeting the overlapping grades C and D.

4.5 Assessment Objectives

The Assessment Objectives describe the intellectual and practical skills which candidates should be able to demonstrate, in the context of the prescribed content. Candidates should demonstrate communication skills, including ICT, using scientific conventions (including chemical equations) and mathematical language (including formulae).

Assessment Objective 1 (AO1): Knowledge and understanding of science and how science works

Candidates should be able to:

- demonstrate knowledge and understanding of the scientific facts, concepts techniques and terminology in the specification;
- show understanding of how scientific evidence is collected and its relationship with scientific explanations and theories;
- show understanding of how scientific knowledge and ideas change over time and how these changes are validated.

Assessment Objective 2 (AO2): Application of skills knowledge and understanding

Candidates should be able to:

- apply concepts, develop arguments or draw conclusions related to familiar and unfamiliar situations;
- plan a scientific task, such as a practical procedure, testing an idea, answering a question or solving a problem;
- show understanding of how decisions about science and technology are made in different situations, including contemporary situations and those raising ethical issues;
- evaluate the impact of scientific developments or processes on individuals, communities or the environment.

Assessment Objective 3 (AO3): Practical, enquiry and data-handling skills

Candidates should be able to:

- carry out practical tasks safely and skillfully;
- evaluate the methods they use when collecting first-hand and secondary data;
- analyse and interpret qualitative and quantitative data from different sources;
- consider the validity and reliability of data in presenting and justifying conclusions.

nd understanding
f knowledge and understanding, analysis and evaluation

tween the components and the assessment objectives of th
vn in the following grid.

Weighting of Assessment Objectives by l		
AO1	AO2	AO3
15%	16.3%	2.0%
13%	18.3%	2.0%
2%	6%	25.3%
30%	40.6%	29.3%

of Written Communication

ected to:

t is legible and that spelling, punctuation and grammar are ;
ar;

ation in a form that suits its purpose;

structure and style of writing.

of written communication will be assessed in the Ideas in C
idy (A329) or Practical Investigation (A330).

5.1 Nature of Skills Assessment

Rationale

The skills assessment accounts for 33.3% of the marks for this specification. There is some choice of the material that is presented for assessment. However it is hoped that candidates have opportunities to develop their skills in all aspects of the tasks described here and then present the highest scoring piece of work.

Skills assessment should arise naturally out of teaching, so that it can be assessed by teachers, internally standardised and then externally moderated by OCR. Candidates are required to submit work for either Unit 4 (A329) or Unit 5 (A330).

Practical Data analysis and Case study (Unit 4, A329)

The Unit 4 skills assessment comprises two elements: the critical analysis of primary data, and a Case Study on a topical (scientific) issue.

First-hand experience of the problems of collecting valid and reliable data can give candidates a better sense of what the difficulties really are, a 'feel' for how great they are in specific cases, and provide a context for beginning to understand how to tackle and perhaps overcome them. Analysis and interpretation of data teaches how scientists use experimental evidence to develop and test theories. Evaluation of procedures and data shows how the reliability of scientific findings can be assessed.

The Case Study is designed to motivate candidates and give them an insight into how science is reported to the public, and how they can explore the validity of underlying research and claims or recommendations based on the research. Centres should note that marks for both elements of Unit 4 (A329) must be submitted in the same examination session.

Element 1: Data Analysis: Marks submitted out of 16

Candidates either singly or collaboratively take part in a practical procedure in order to collect primary data. Candidates are assessed on their ability to analyse and evaluate the data collected and the limitations of the techniques used. It is not essential for candidates to collect all of the data which is to be used in this exercise. Their own first hand data may be supplemented with extra data from other candidates or classes, demonstrations or other sources.

Marks are awarded for two strands, Interpretation (Strand I) and Evaluation (Strand E). The two marks which make up the assessment total for this element of skills assessment must both come from the same activity.

Element 2: Case Study Marks submitted out of 24

This assignment should arise naturally from work on the course or from an issue that arises while candidates are following the course. It should be related to an aspect of science that involves an element of controversy, in terms either of the interpretation of evidence, or of the acceptability of some new development. Topics for study should be selected by candidates in discussion with teachers, and should be seen as an extension or consolidation of studies undertaken as a normal part of the course. The work should be capable of being completed within approximately 4-6 hours

over a period of time, for example, one lesson per week for half a term, with some non-contact time.

Practical Investigation (Unit 5, A330)

The use of practical investigations to assess skills in science was based on research in a number of centres, particularly the University of Durham. For more than 10 years, it has formed the basis of coursework assessment for National Curriculum science.

Investigations require the drawing together of skills in planning, collecting data, interpreting data and evaluation. They provide an effective and valid assessment instrument for a course which is seen as a basis for further studies and possible future careers in science. However, the regulations used at Key Stage 4 over the past 5-year cycle have been constructed in a way which has restricted the variety of work attempted and has led to rather mechanical 'criterion matching', rather than genuine open-ended work.

For this specification, the basic structure of investigations is retained, but the emphasis on prediction is removed, allowing a much wider range of activities and approaches. A different marking style has been developed, drawing more on the professional judgment of teachers.

The task aims to motivate candidates and help them to appreciate the importance of having a clear and manageable question, to learn how to choose equipment and use it appropriately, and to design suitable apparatus for making observations and measurements. First-hand experience of the problems of collecting valid and reliable data can give candidates a better sense of what the difficulties really are, and a 'feel' for how great they are in specific cases, and provide a context for beginning to understand how to tackle and perhaps overcome these.

Candidates are required to complete one single practical investigation. The Investigation, accounts for 33% of the marks for this specification. It is assessed by teachers, internally standardised, and then externally moderated.

Within this science suite, investigative work is designed to have a broad definition. In addition to confirming the predicted effect of a variable on a system over a range, the definition also includes more speculative investigation of systems where no clear prediction can be made in advance, e.g. where there is little relevant explanatory theory available in the course, or where the experimental material is likely to be variable, for example in surveys of distribution of species. It also includes tasks which involve determining the consistency of measurements e.g. comparing the characteristics of different artefacts, obtaining evidence for the 'normal' variation in respiratory peak flow-rates of an individual, etc.

The initial stimulus for an investigation should arise from class teaching or discussion which ensures that candidates are aware of suitable practical techniques and have some relevant background theoretical knowledge.

This element of the assessment is based on complete, first hand practical investigations. Candidates may complete as many investigations as they wish during the course. The final mark will be the total for the highest-scoring single piece of work assessed. It is not permitted to aggregate together marks taken from different investigations. Where appropriate, first hand data collected by the candidate may be supplemented by secondary data from other sources. In such cases, credit for collecting data should be based on the overall quality of all the data obtained or selected.

Marks are awarded for 5 strands of the investigation, with each strand marked on a scale of 0–8.

5.2 Marking Internally Assessed Work

Arrival at Strand Marks

The method of marking the skills assessment is the same across this Science suite.

The award of marks is based on the professional judgement of the science teacher, working within a framework of descriptions of performance. Within each strand, each line in the marking grids represents a different aspect of performance. For each of these, a series of four descriptions of performance illustrates what might be expected for candidates working at different levels.

Marking decisions should be recorded on marking grids. A master copy is provided in the skills assessment guidance booklet. The completed grid serves as a cover-sheet for the work if it is required for moderation.

Candidates may not always report their work in a particular order. So, evidence of achievement in a strand may be located almost anywhere in the work. Thus, it is necessary to look at the whole piece of work for evidence of each strand in turn.

Within any one strand, each aspect should be considered in turn. A tick on the grid should be used to indicate the performance statement that best matches the work.

Where the maximum mark is 8, intermediate marks 1, 3, 5 or 7 can be used where performance exceeds that required by one statement, but does not adequately match that required by the next higher statement (e.g. if the work significantly exceeds what is required for 4 marks, but does not reach the standard for 6, then the tick should be placed on the dividing line between the 4 and 6 mark boxes).

Where a decision is based partly on the teacher's observation of the candidate at work, the work should be annotated to record this at an appropriate point.

In some cases, in order to allow credit for the widest possible variety of activities, an aspect of performance is represented by two (or more) rows of mark descriptors. In such cases, where a row is not relevant or appropriate for a particular activity, it should be left blank and excluded from the 'best-fit' marking judgement and the more appropriate alternative row used.

When each aspect of the performance within a strand have been assessed in this way, the pattern of achievement is interpreted by a 'best-fit' judgement to give a mark for that strand.

This method of marking can be applied even where there is a wide variation between performance in different aspects. Thus, weak performance in one aspect need not depress marks too far if other aspects show better performance.

Recording and submitting marks

Skills Assessment Forms will be provided for centres to record marks submitted for moderation. The final mark should be submitted to OCR on form MS1 by **15th May** in the year of entry. These forms are produced and dispatched at the relevant time, based on entry information provided by the Centre.

All assessed work which has contributed to candidates' final totals must be available for moderation.

ids in this element; Interpreting Data and Evaluation. The descriptors for each to those found in Unit 5 Practical Investigation (A330).

ing Data

ected to be able to:

cess a set of data in such a manner as to bring out any 'patterns'¹ that are 4, 2.1, 2.3-4);

ns based on these patterns (laS 2.4);

clusions to scientific theories or understanding (laS 3.1, 3.3, 3.4).

le, each row represents increasing achievement in a different aspect of

Strand I Mark			
2	4	6	8
Display limited numbers of results in tables, charts or graphs, using given axes and scales.	Construct simple charts or graphs to display data in an appropriate way, allowing some errors in scaling or plotting.	Correctly select scales and axes and plot data for a graph, including an appropriate line (normally a line of best fit) or construct complex charts or diagrams (e.g. stacked histograms, species distribution maps).	Additionally, indicate the spread of data (e.g. through scatter-graphs or error bars) and give clear keys for displays involving multiple data sets.
Select individual results as a basis for conclusions.	Carry out simple calculations (e.g. correct calculation of averages from repeated readings).	Use mathematical comparisons between results to support a conclusion.	Use complex processing to reveal patterns in the data (e.g. statistical methods, use of inverse relationships or calculation of gradient of graphs).
Note differences between situations/cases, or compare individual results.	Identify trends or general correlations in the data.	Describe formal or statistical relationships within the cases/situations studied.	Review the extent of, or limitations to, formal conclusions in relation to the scatter evident in the data.
Link the outcomes to previous experience or 'common sense'.	Relate the conclusion to scientific ideas/ explanations.	Justify the conclusion by reference to relevant scientific knowledge and understanding.	Use detailed scientific knowledge to explain all aspects of the given conclusion.

ans similarities, or differences, or the presence or absence of a relationship (e.g. a a factor and an outcome, or a trend linking two variables)

ected to be able to look back at the experiment they have carried out, showing
 rned from doing it and explaining how they would modify it in the light of this,
 t out again. These suggestions may demonstrate understanding of:

ollecting valid and reliable data (laS 1.1–3);

the design of the data set collected, such as imperfect control of other
 e size and matching of samples compared (laS 2.3, 2.6–7);

level of confidence that can be placed in these conclusions (laS 2.2–3, 2.6–7).

le, each row represents increasing achievement in a different aspect of

Strand E Mark			
2	4	6	8
Make a relevant comment about how the data was collected and safety procedures.	Comment on the limitations to accuracy or range of data imposed by the techniques and equipment used.	Suggest improvements to apparatus or techniques, or alternative ways to collect the data, but without sufficient practical detail.	Describe in detail improvements to the apparatus or techniques, or alternative ways to collect the data, and explain why they would be an improvement.
Make a claim for accuracy or reliability, but without appropriate reference to the data.	Note the presence or absence of results that are beyond the range of experimental error.	Use the general pattern of results or degree or scatter between repeats as a basis for assessing accuracy and reliability.	Consider critically the reliability of the evidence, accounting for any anomalies.
Relate judgement of the reliability (or otherwise) of the conclusions only to techniques used, not to data collected.	Link confidence in the conclusion to the apparent reliability of the data collected.	Discuss the precision of apparatus and techniques, the range covered and reliability of data to establish a level of confidence in the conclusions.	Identify weaknesses in the data and give a detailed explanation of what further data would help to make the conclusion more secure.

In everyday life, citizens most often become aware of science-related issues through reports in the media: newspapers, teenage magazines, television, etc. This element of the assessment is designed to help candidates develop strategies for evaluating such information, and to increase awareness of appropriate ways of making decisions about such issues.

Ideally, the study should arise from such a media source. Suitable topics involve some degree of controversy, or disagreement, either about the interpretation of the scientific evidence, or about how individuals or society should respond. The title for a Case Study is best phrased as a question to be answered by careful balancing of evidence and opinions from a variety of sources.

Suitable topics often fall into one of three main types:

- Evaluating claims where there is uncertainty in scientific knowledge (e.g. “Is there life elsewhere in the Solar System?” or “Does using mobile phones cause risk of brain damage?”). Controversies of this type focus attention on the relationship between data and explanations in science, and on the quality of research which underlies competing claims.
- Contributing to decision making on a science-related issue (e.g. “Should the government restrict research into human cloning?”). Studies in this category are more likely to involve elements of personal choice, values and beliefs, and may involve balancing of risks and benefits of any proposed action.
- Personal or social choices (e.g. “Should my child receive the triple MMR vaccine?”). Ethical and personal issues are likely to figure in such studies, but it is important to evaluate these in relation to what is known about the science which underlies the issue.

In all cases, an important factor in choice of subject should be the availability of information giving a variety of views in forms that can be accessed by the candidate. Candidates may be provided with the initial stimulus for the study, but should be encouraged to search for a range of opinions in order to reach a balanced conclusion.

The subject need not be restricted to topics studied in the course. However, it is necessary for the candidate to apply some relevant scientific knowledge and understanding to discussion of the issues raised. This is most likely to be the case if the study arises naturally during normal work on the course.

Candidates need not all study the same, or related, topics. Motivation is greatest if they are given some degree of autonomy in the choice of topic. This may be achieved by allowing choice of different issues related to a general topic (e.g. different aspects of air pollution when studying Air Quality) or by encouraging candidates to identify topics of interest and begin collecting resource materials over an extended period. At a time chosen by the centre, candidates then complete their Case Study, and may each be working on different topics.

Candidates will find it helpful to have a clear sense of audience in their writing – perhaps candidates in year 9, to encourage them to explain the basic science behind the topic.

The Case Study will often take the form of a 'formal' written report. However, candidates should not be discouraged from other styles of presentation, for example:

- a newspaper or magazine article;
- a PowerPoint presentation;
- a poster or booklet;
- a teaching/learning activity such as a game;
- a script for a radio programme or play.

In all cases, sufficient detail must be included to allow evaluation in all of the performance areas. Some types of presentation would require supporting notes.

A Case Study represents a major piece of work and it is not expected that candidates will attempt more than one during the two years of the course. If a candidate has attempted more than one case study, then the total for the assessment should be the highest total for any one case study.

It is **not** permitted to aggregate marks from two or more different pieces of work, nor to add marks obtained from separate, limited range tasks, exercises or part-studies.

Marking Criteria – Case Study

Marks are awarded under four headings, A, B, C and D.

Because of the risk of some studies becoming excessively long, it is important to link marks to the quality of the work done, rather than the quantity.

The four areas of performance to be awarded credit are:

A: Quality of selection and use of information, on a scale of 0–4marks

Here candidates should show an awareness of sources of information such as their own notes, text books or encyclopedias, or the internet. They should consider the reliability of any sources used. All sources should be credited, and it should be clear where each piece of information has come from. Credit is given for being selective in choosing only relevant material. Direct quotations should be acknowledged.

B: Quality of understanding of the case, on a scale of 0–8 marks

Candidates should describe the basic science which helps understanding of the topic, and apply it to evaluate the reliability of claims made. In many cases, they may follow a topic beyond the normal limits of the specification, and credit should be awarded for understanding whether within or beyond the specification.

C: Quality of conclusions, on a scale of 0–8 marks

Different evidence, arguments or views should be compared and evaluated and used as a basis for a balanced conclusion or proposal for action.

D: Quality of presentation, on a scale of 0–4 marks

Communication skills should be rewarded for effective presentation including use of different forms for presenting different types of information (e.g. pictures, tables, charts, graphs, etc.).

ect and organise information from a variety of sources, bearing in mind both
 idy and the apparent reliability of the sources. It is expected that centres will
 sic selection of resources available for candidates to work from. A survey of the
 e course will identify topics which are likely to be relevant, topical and of interest
 addition to standard textbooks and library books, resources are available from
 onmental groups and in popular science magazines, as well as through the

be encouraged to seek out their own additional resources, but should not be
 ent on this, and in particular, should not be dependent on home or out of school

for selection of appropriate material from the available resources, rather than
 ing. It will also be given for judgement shown in selecting from a variety of
 alanced view of the topic. Good work is characterised by the ability of the
 and re-structure information to suit the purpose of the study.

ididates may wish to collect information about the public acceptability of an idea
 k through questionnaires (administered to classmates or other groups) or to test
 gh experimental work. Whilst relevant work of these types may be credited, it
 e the study.

ates should record the sources of information they have used. The assignment
 introduction to the value of crediting sources in scientific communication.

Strand A Mark			
1	2	3	4
Very little information is given beyond that provided by the original stimulus material.	Information from a limited range of additional sources is included, although some may be irrelevant or inappropriate to the study.	Relevant information is selected from a variety of sources.	Sources of information are assessed for reliability as a basis for selection of relevant information from a wide variety of sources.
	Sources are identified by incomplete or inadequate references.	References to sources are clear, but limited in detail.	References to these sources are clear and fully detailed.
	Direct quotations are rarely indicated as such.	Direct quotations are generally acknowledged.	The sources of particular opinions are indicated at appropriate points in the text of the report.

Candidates should make reference to explanatory scientific theory to help them understand the significance of the information they are dealing with. However, controversies in science are often found in areas where there is no (GCSE level) descriptive theory to provide a basis for understanding and evaluating the issues involved. In such cases, candidates should draw on their own knowledge, especially IAS 2 (Correlation and cause) to justify the conclusions they reach on the basis of the data they have collected.

Candidates should not be used to extend or assess the candidate's knowledge of basic science unrelated to the topic, but rather to encourage them to see how the science available can be related to topical issues to help them reach valid judgements. Some candidates may wish to go beyond what they have been taught in class and, if they find and use relevant theory which is directly relevant to the Case Study, this can help to raise their mark. However, marks should not be given to uncritical copying of large amounts of theory from texts.

Candidates should provide some background to the case study in relation to relevant scientific theory and should also evaluate how well-founded are links between the available evidence and the conclusions made on the basis of the evidence. Where little explanatory theory is available at GCSE level, candidates should draw on Ideas about Science 2, 3 and 4 to help them evaluate the case study.

The quality of work depends on understanding of:

Science 1: Data and its limitations (mainly 1.2, 1.3 and 1.4);

Science 2: Correlation and cause (mainly 2.1, 2.2, 2.4, - 2.7).

Strand B Mark			
2	4	6	8
Only superficial mentions of science explanations, often not correctly applied to the case.	Provides a basic outline of the main scientific ideas which are relevant to the case.	Provides a detailed review of the scientific knowledge needed to understand the issues studied.	Considers how different views described in the study can be supported by detailed scientific explanations.
Sources are uncritically quoted without distinguishing between scientific evidence and unsupported claims.	Science content and data in sources is recognised.	Claims and opinions are linked to the scientific evidence they are based on.	The quality of scientific evidence in sources is evaluated in relation to the reliability of any claims made.

an account of different views or opinions which are represented in the
 . Credit will be given for discussion of the perceived benefits and associated
 d actions, and for judgements of the acceptability of any conclusions reached.

ould be such that there is scope for taking views about the acceptability of
 of action.

of the Case Study will be linked to understanding of:

ence 1: Data and its limitations (mainly parts 1.2 and 1.4);

ence 5: Risk (mainly parts 5.1, 5.2, 5.4, 5.6 and 5.7);

ence 6: Making decisions about science and technology (mainly parts 6.3, 6.4,

scriptions represents a different aspect of the use of resources. For each row,
 ent displayed should be indicated, then a single overall mark on a scale of 0–
 ld be awarded, using professional judgement to decide the best overall match
 hown.

Strand C Mark			
2	4	6	8
Information is unselectively reported without taking any clear view about any course of action.	Claims for a particular idea, development or course of action are reported without critical comment.	Claims and arguments for and against are reported, but with little attempt to compare or evaluate them.	Details of opposing views are evaluated and critically compared.
A conclusion is stated without reference to supporting evidence.	A conclusion is based on evidence for one view only.	Some limits or objections to the conclusion are acknowledged.	Alternative conclusions are considered, showing awareness that different interpretations of evidence may be possible.

be encouraged to be creative and imaginative in their choice of method and communicating their findings. The report may be in a variety of forms, including formal newspaper articles for an identified public audience, PowerPoint presentations, signs, scripts for a radio programme or play etc. Whatever form of presentation is used, it should be supported by sufficient documentation to allow assessment of all four criteria. It should also be remembered that the work may need to be posted to a moderator for the course. Where electronic media are included, a paper print-out must be included for marking purposes.

Effort and fitness for purpose should be rewarded in the assessment, rather than the quality of the work.

When suggestions are given, candidates should be encouraged to structure the report clearly. This helps to improve motivation and make the work "special", thinking about a good title. Contents can help candidates to organise their ideas. Use of tables of contents, headings and links between sections of text are valuable in this context.

Visual aids should be used where they lead to clearer communication of ideas. These may be hand-drawn leaflets or 'clip-art' sources, or drawn by candidates: they may be pictorial or diagrams. Charts and graphs should be used to present and summarise data. Reports may be handwritten or word-processed.

Candidates should be encouraged to think carefully of their target audience and how to communicate their ideas clearly.

Strand D Mark			
1	2	3	4
The report has little or no structure or coherence, or follows a pattern provided by worksheets.	The report has an appropriate sequence or structure.	Information is organised for effective communication of ideas, with contents listing, page numbering etc. as appropriate to aid location of key elements.	Considerable care has been taken to match presentation and format to present issues and conclusions clearly and effectively to a chosen audience.
There is little or no visual material (charts, graphs, pictures etc.) to support the text.	Visual material is merely decorative, rather than informative.	Visual material is used to convey information or illustrate concepts.	Pictures, diagrams, charts and or tables are used appropriately and effectively to convey information or illustrate concepts.
Spelling, punctuation and grammar are of generally poor quality, with little or no use of appropriate technical or scientific	Spelling, punctuation and grammar are of variable quality, with limited use of appropriate technical or scientific	Spelling, punctuation and grammar are generally sound, with adequate use of appropriate technical or scientific	The report is concise, with full and effective use of relevant scientific terminology. Spelling, punctuation and grammar are

Unit 5 (A330), Practical Investigation

This unit is designed to test the ability of the candidate to plan and undertake a whole investigation or problem-solving task. Scoring individual marks in different tasks, or parts of tasks, removes this holistic element, and can result in performances of very different quality leading to the same final assessment total. For this reason, the final unit mark for each candidate will be the highest total mark achieved on any one task. This total mark is obtained by adding together the marks achieved on each strand of the work on that task.

The requirement is for the highest mark from a single piece of work. It is not essential for this to be complete, in the sense of providing evidence across all strands. It may happen that some candidates achieve their highest total for a piece of work in which evidence for one or more strands is missing; in such cases this total should be chosen as the final assessment total.

Centres may assess the performance of candidates on any occasion when investigative work is taking place throughout the course.

ions are likely to arise out of work on most or all of the course modules. It should be suggested to candidates, but they should also have opportunities to modify to suggest questions or tasks to investigate in topic areas they are studying. (Candidates should) obtain more credit for tackling somewhat more demanding tasks, and in devising the question/task, rather than 'playing safe' with a given, or routine task requiring little skill in the use of equipment.

Candidates should be encouraged to plan an investigation before starting, there is limited time for them to produce a detailed written plan – as their actions should be open to them to proceed. Indeed, it is good practice to try taking a few measurements or variations to get a 'feel' for the equipment and the system being investigated, and to develop a detailed data collection strategy. For that reason, the candidate's understanding of the system and the quality of data is better assessed from the final data set they present (see Strand C) than from an initial plan.

The quality of strand S focuses on:

the demand of the task and approach chosen;

the equipment, materials and techniques;

the independence shown in formulating the task and the approach to it.

Strand S Mark			
2	4	6	8
Simple measurement or comparison task, based on straightforward use of simple equipment	Routine task requiring only limited precision or range of data to be collected.	Straightforward task of limited complexity, but requiring good precision or a wide range of data.	Complex task requiring high levels of precision/reliability in the data collected.
Follow a given technique, but with very limited precision or reliability.	Select and use basic equipment to collect a limited amount of data.	Select and use techniques and equipment which are appropriate for the range of data required.	Justify the choice of equipment and technique to achieve data which is precise and reliable.
The task has been set by the teacher and/or is based on specific, task-related structured worksheets.	The task is closely defined by the teacher, but is carried out with little further guidance.	The task is defined by the candidate from a more general brief, then carried out independently.	The topic is reviewed by the candidate to justify a choice of task. The work is completed independently.

ected to be able to collect a set of data in a manner which
ow to ensure (and assess) quality.

a set depends on:

individual data points, which in turn depends on:

carefully the measurements have been taken, and how ac-
uments are (laS 1.1-2);

much variation or scatter there is in repeated measuremer
e been taken to assess and deal with this (laS 1.1–4);

ther the instruments used, or the way they are used, results
differ from the ‘true’ value of the quantity (laS 1.1–2);

design of the set of data points collected, that is:

ther enough data points have been collected (laS 2.1, 2.3,

ther these cover an adequate range (of cases, or situations
pendent variable) (laS 2.3);

relationship is being explored) whether the design of the dæ
t of other variables to be excluded (for example (laS 2.2–3

use preliminary experiments or other information to confirm
ge of values to be tested will lead to results of good quality

e written to refer to primary data that the candidate has colle
from secondary sources, the statements should be read as
sed to ‘collected’). The mark awarded should be based on
ole.

Strand C Mark		
2	4	6
Little or no care has been taken to identify or control outside influences.	Identifies some factors which may affect the outcomes and need to be controlled or accounted for.	Identifies the majority of factors which may affect the outcomes and need to be controlled or accounted for.
The data is very limited in amount (e.g. isolated individual data points, with no clear pattern), covering only part of the range of relevant cases/ situations, with no checking for reliability.	An adequate amount or range of data is collected, but with little or no checking for reliability.	Data is collected to cover the range of relevant cases/ situations, with regular repeats or checks for reliability.
Little care evident in use of apparatus. Data generally of low quality.	Use of techniques and apparatus generally satisfactory. Data of variable quality. with	Sound techniques in use of apparatus/ equipment. Data of generally good quality.

ected to be able to:

ress a set of data in such a manner as to bring out any ‘patterns’² that are
4, 2.1, 2.3–4);

ns based on these patterns (laS 2.4);

clusions to scientific theories or understanding (laS 3.1, 3.4, 3.5).

Strand I Mark			
2	4	6	8
Display limited numbers of results in tables, charts or graphs, using given axes and scales.	Construct simple charts or graphs to display data in an appropriate way, allowing some errors in scaling or plotting.	Correctly select scales and axes and plot data for a graph, including an appropriate line (normally a line of best fit) or construct complex charts or diagrams (e.g. stacked histograms, species distribution maps).	Additionally, indicate the spread of data (e.g. through scatter-graphs or error bars) and give clear keys for displays involving multiple data sets.
Select individual results as a basis for conclusions.	Carry out simple calculations (e.g. correct calculation of averages from repeated readings).	Use mathematical comparisons between results to support a conclusion.	Use complex processing to reveal patterns in the data (e.g. statistical methods, use of inverse relationships or calculation of gradient of graphs).
Note differences between situations/cases, or compare individual results.	Identify trends or general correlations in the data.	Describe formal or statistical relationships within the cases/situations studied.	Review the extent of, or limitations to, formal conclusions in relation to the scatter evident in the data.
Link the outcomes to previous experience or ‘common sense’.	Relate the conclusion to scientific ideas/explanations.	Justify the conclusion by reference to relevant scientific knowledge and understanding.	Use detailed scientific knowledge to explain all aspects of the given conclusion.

ans similarities, or differences, or the presence or absence of a relationship (e.g. a
a factor and an outcome, or a trend linking two variables)

ected to be able to look back at the investigation they have carried out, have learned from doing it and explaining how they would modify it in the light to carry it out again. These suggestions may demonstrate understanding of:

ollecting valid and reliable data (laS 1.1–2);

the design of the data set collected, such as imperfect control of other e size and matching of samples compared (laS 2.3, 2.6–7);

level of confidence that can be placed in these conclusions (laS 2.2-3, 2.7–8).

Strand E Mark			
2	4	6	8
Make a relevant comment about how the data was collected and safety procedures.	Comment on the limitations to accuracy or range of data imposed by the techniques and equipment used.	Suggest improvements to apparatus or techniques, or alternative ways to collect the data, but without sufficient practical detail.	Describe in detail improvements to the apparatus or techniques, or alternative ways to collect the data, and explain why they would be an improvement.
Make a claim for accuracy or reliability, but without appropriate reference to the data.	Note the presence or absence of results that are beyond the range of experimental error.	Use the general pattern of results or degree or scatter between repeats as a basis for assessing accuracy and reliability.	Consider critically the reliability of the evidence, accounting for any anomalies.
Relate judgement of the reliability (or otherwise) of the conclusions only to techniques used, not to data collected.	Link confidence in the conclusion to the apparent reliability of the data collected.	Discuss the precision of apparatus and techniques, the range covered and reliability of data to establish a level of confidence in the conclusions.	Identify weaknesses in the data and give a detailed explanation of what further data would help to make the conclusion more secure.

t clearly and effectively on one's work is essential in order to demonstrate
 ie Ideas about Science that relate to practical investigations.

or three aspects of reporting and communicating a practical investigation:

of the report, with all practical procedures clearly described, all parameters and
 ted, a full analysis of the evidence, and an evaluation of both procedures and

f the report, including layout and effective sequencing, use of illustrations as
 d use of graphs and charts to present information;

English, including accurate grammar, punctuation and appropriate use of
 i.

Strand P Mark			
2	4	6	8
The purpose/ context of the investigation is not made clear. Key features of experi- mental procedures are omitted or unclear.	The purpose of the work is stated. Main features of the work are described, but there is a lack of detail.	There is a clear statement of the question/task and its scope. Practical procedures are clearly described.	All aspects of the task are reviewed. Practical procedures are discussed critically and in detail.
Major experimental parameters are not recorded. Some data may be missing.	Most relevant data is recorded, but where repeats have been used, average values rather than raw data may be recorded.	All raw data, including repeat values, are recorded.	All relevant parameters and raw data including repeat values are recorded to an appropriate degree of accuracy.
Labelling of tables is inadequate. Most units are absent or incorrect.	Labelling is unclear or incomplete. Some units may be absent or incorrect.	All quantities are identified, but some units may be omitted.	A substantial body of information is correctly recorded to an appropriate level of accuracy in well-organised ways.
Observations are incomplete or sketchily recorded.	Recording of observations is adequate but lacks detail.	Observations are adequate and clearly recorded.	Observations are thorough and recorded in full detail.
Spelling, punctuation and grammar are of generally poor quality. Little or no relevant technical or scientific vocabulary is used.	Use of appropriate vocabulary is limited. Spelling, punctuation and grammar are of very variable quality.	Appropriate scientific vocabulary is used. Spelling, punctuation and grammar are generally sound.	There is full and effective use of relevant scientific terminology. Spelling, punctuation and grammar are almost faultless.

5.3 Regulations for Internally Assessed Work

Supervision and authentication of work

OCR expects teachers to supervise and guide candidates who are undertaking work that is internally assessed. The degree of teacher guidance will vary according to the kind of work being undertaken. It should be remembered, however, that candidates are required to reach their own judgements and conclusions.

When supervising internally assessed tasks, teachers are expected to:

- offer candidates advice about how best to approach such tasks;
- exercise supervision of work in order to monitor progress and to prevent plagiarism;
- ensure that the work is completed in accordance with the specification requirements and can be assessed in accordance with the specified mark descriptions and procedures.

Coursework should, wherever possible, be carried out under supervision. However, it is accepted that some tasks may require candidates to undertake work outside the Centre. Where this is the case, the Centre must ensure that sufficient supervised work takes place to allow the teachers concerned to authenticate each candidate's work with confidence.

Production and presentation of internally assessed work

Candidates must observe certain procedures in the production of internally assessed work.

- Any copied material must be suitably acknowledged.
- Where work includes secondary data, the original sources must be clearly identified.
- Each candidate's assessed work submitted for moderation should be stapled together at the top left hand corner and have a completed cover sheet as the first page.

Annotation of candidates' work

Each piece of assessed work should be annotated to show how the marks have been awarded in relation to the mark descriptions.

The writing of comments on candidates' work provides a means of dialogue and feedback between teacher and candidate and a means of communication between teachers during internal standardisation of coursework.

However, the main purpose of annotating candidates' coursework is to provide a means of communication between teacher and moderator, showing where marks have been awarded and why they have been awarded.

Annotations should be made at appropriate points in the margins of the script of all work submitted for moderation. The annotations should indicate where achievement for a particular skill has been recognised.

It is suggested that the minimum which is necessary is that the 'shorthand' mark descriptions (for example, Ea8) should be written at the point on the script where it is judged that the work has met the mark description.

Moderation

All internally assessed work is marked by the teacher and internally standardised by the Centre. Marks are then submitted to OCR by a specified date, after which moderation takes place in accordance with OCR procedures. The purpose of moderation is to ensure that the standard of the award of marks is the same for each Centre and that each teacher has applied the standards appropriately across the range of candidates within the Centre.

It is the responsibility of the Centre to carry out effective internal standardisation to ensure that similar standards are applied by each teacher involved in the assessment. The Moderator will require a written statement describing how internal standardisation has been carried out within the Centre.

External moderation will be by postal sample selected by the Moderator.

The sample will represent performance across the whole ability range from the Centre. The sample of work which is presented to the Moderator for moderation must show how the marks have been awarded in relation to the mark descriptions.

Separate cover sheets are required for each candidate's work in the sample submitted for moderation.

Minimum requirements for internally assessed work

If a candidate submits no work for this internally assessed unit, then the candidate should be indicated as being absent from that unit on the mark sheets submitted to OCR. If a candidate completes any work at all for an internally assessed unit, then the work should be assessed according to the criteria and mark descriptions and the appropriate mark awarded, which may be zero.

King Unit Entries

that centres must be registered with OCR in order to make any entries, including tries. It is recommended that centres apply to OCR to become a registered centre of making their first entries. Centres should be aware that a minimum of ten or summer examinations is normally required.

Options

A321, A322 and A323 candidates must be entered for either the Foundation Tier or er option. It is not necessary for candidates to enter at the same tier in every unit. nay, if they wish, attempt papers at both tiers, but not in the same examination e the papers will be timetabled simultaneously.

Option code	Component to be taken
F	01 Chemistry A Unit 1 – modules C1, C2, C3 Foundation
H	02 Chemistry A Unit 1 – modules C1, C2, C3 Higher
F	01 Chemistry A Unit 2 – modules C4, C5, C6 Foundation
H	02 Chemistry A Unit 2 – modules C4, C5, C6 Higher
F	01 Chemistry A Unit 3 – Ideas in Context plus C7 Foundation
H	02 Chemistry A Unit 3 – Ideas in Context plus C7 Higher
–	01 Chemistry A Unit 4 – Practical Data Analysis and Case Study
–	01 Chemistry A Unit 5 – Practical Investigation

tries must be made by 21 October for the January session and by 21 February for the l.

King Qualification Entries

must be entered for certification code **J634** to claim their overall GCSE grade.

tion entry is not made, no overall grade can be awarded.

who has completed all the units required for the qualification may enter for certification same examination session (within a specified period after publication of results) or at a .

tion will be available in June 2008 and every January and June thereafter.

cannot be declined.

6.3 Grading

GCSE results are awarded on the scale A*-G. Units are awarded a* to g. Grades are awarded on certificates. Results for candidates who fail to achieve the minimum grade (G or g) will be recorded as unclassified (U or u).

In modular schemes candidates can take units across several different sessions. They can also re-sit units or choose from optional units available. When working out candidates' overall grades OCR needs to be able to compare performance on the same unit in different sessions when different grade boundaries have been set, and between different units. OCR uses uniform marks to enable this to be done.

A candidate's uniform mark is calculated from the candidate's raw mark. The raw grade boundary marks are converted to the equivalent uniform mark boundary. Marks between grade boundaries are converted on a pro rata basis.

When unit results are issued, the candidate's unit grade and uniform mark are given. The uniform mark is shown out of the maximum uniform mark for the unit e.g. 41/50.

Results for each unit will be published in the form of uniform marks according to the following scales.

	Unit Grade								
	a*	a	b	c	d	e	f	g	u
Units 1 and 2	50-45	44-40	39-35	34-30	29-25	24-20	19-15	14-10	10-0
Units 3, 4 and 5	100-90	89-80	79-70	69-60	59-50	49-50	39-30	29-20	19-0

Higher tier candidates may achieve an "allowed e". Higher tier candidates who miss a grade 'e' will be given a uniform mark in the range f-u but will be graded as 'u'.

Candidates' uniform marks for each module are aggregated and grades for the specification are generated on the following scale.

Qualification Grade								
A*	A	B	C	D	E	F	G	U
300-270	269-240	239-210	209-180	179-150	149-120	119-90	89-60	59-0

The candidate's grade will be determined by this total mark. Thus, the grade awarded will depend in practice upon the extent to which the candidate has met the assessment objectives overall. Shortcomings in some aspects of the assessment may be balanced by better performance in others. Candidates achieving less than the minimum mark for grade G will be unclassified.

6.4 Result Enquiries and Appeals

Under certain circumstances, a centre may wish to query the grade available to one or more candidates or to submit an appeal against an outcome of such an enquiry. Enquiries about unit results must be made immediately following the series in which the relevant unit was taken.

For procedures relating to enquires on results and appeals, centres should consult the OCR *Handbook for Centres* and the document *Enquiries about Results and Appeals – Information and Guidance for Centres* produced by the Joint Council. Copies of the most recent editions of these papers can be obtained from OCR.

6.5 Shelf-Life of Units

Individual unit results, prior to certification of the qualification, have a shelf-life limited only by that of the qualification.

6.6 Unit and Qualification Re-sits

Candidates may re-sit any unit an **unlimited** number of times.

For each unit the best score will be used towards the final overall grade.

Candidates may enter for the full qualification an unlimited number of times.

6.7 Guided Learning Hours

GCSE Chemistry A requires 120 guided learning hours in total.

6.8 Code of Practice/Subject Criteria/Common Criteria Requirements

These specifications comply in all respects with the revised *GCSE, GCE, VCE, GNVQ and AEA Code of Practice 2005/6*, the subject criteria for GCSE Chemistry A and *The Statutory Regulation of External Qualifications 2004*.

6.9 Arrangements for Candidates with Particular Requirements

For candidates who are unable to complete the full assessment or whose performance may be adversely affected through no fault of their own, teachers should consult the Access Arrangements and Special Consideration Regulations and Guidance Relating to Candidates who are Eligible for Adjustments in Examinations. In such cases advice should be sought from OCR as early as possible during the course.

6.10 Prohibited Qualifications and Classification Code

Every specification is assigned to a national classification code indicating the subject area to which it belongs.

Centres should be aware that candidates who enter for more than one GCSE qualification with the same classification code will have only one grade (the highest) counted for the purpose of the School and College Performance Tables.

The classification code for this specification is 1110.

7.1 Overlap with other Qualifications

This specification has been developed alongside GCSE Science A and GCSE Additional Applied Science.

7.2 Progression from these Qualifications

GCSE qualifications are general qualifications which enable candidates to progress either directly to employment, or to proceed to further qualifications.

Many candidates who enter employment with one or more GCSEs would undertake training or further part-time study with the support of their employers.

Progression to further study from GCSE will depend upon the number and nature of the grades achieved. Broadly, candidates who are awarded mainly grades G to D at GCSE could either strengthen their base through further study of qualifications at Level 1 within the National Qualifications Framework or could proceed to Level 2. Candidates who are awarded mainly grades C to A* at GCSE would be well prepared to broaden their base through further study of qualifications at Level 2 or to proceed to appropriate qualifications at Level 3 within the National Qualifications Framework.

Candidates intending to proceed to qualifications in chemistry at Advanced Level (Level 3 in the National Qualifications Framework) should have completed either a course in GCSE Chemistry A or both GCSE Science and GCSE Additional Science.

7.3 ICT

In order to move on to more advanced study of science, candidates need to be confident and effective users of ICT. This specification provides candidates with a wide range of appropriate opportunities to use ICT in order to further their study of science.

Opportunities for ICT include:

- using of videos clips to show provide the context for topics studied and to illustrate the practical importance of the scientific ideas;
- gathering information from the Internet and CD-ROMs;
- gathering data using sensors linked to data-loggers or directly to computers;
- using spreadsheets and other software to process data;
- using animations and simulations to visualise scientific ideas;

7.4 Citizenship

Since September 2002, the National Curriculum for England at Key Stage 4 has included a mandatory programme of study for Citizenship.

Core science is designed as a science education for future citizens which not only covers aspects of the Citizenship programme of study but also extends beyond that programme by dealing with important aspects of science which all people encounter in their everyday lives.

Citizenship Programme of Study	Opportunities for Teaching the Issues during the Course
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